

PSYC 270.002: Research Methods in Psychology

Spring 2025 Syllabus

Course Information

Time: Monday, Wednesday, Friday from 9:05-9:55am (Section 001) or 10:10-11:00am (Section 002)

Location: Chapman Hall Room 201 (Section 001) or Carroll Hall Room 111 (Section 002)



Instructor: Dr. Natasha Parikh (she/her/hers)

Feel free to call me Professor/Prof./Dr. P./Parikh, pronounced Puh-reek. My personal favorite is Professor P., for the way it makes me sound like an evil scientist 😊)

Contact: nparikh@unc.edu. If you email me, I will guarantee a response within 1 business day unless otherwise specified. However, if you wait until 11pm the night before an assignment is due and then find you need my help, you may not be able to receive it in time.

Student hours: You have multiple options to come chat with me! These timeslots are a great way to go over topics from class, get advice on your project, or just chat. Feel free to hop in just to say hello!

- Virtual hours held on Zoom (ID: profparikh) on Mondays from 2-3pm
- In person in Davie Hall room 238 on Wednesdays from 12-1pm
- By appointment at <https://calendly.com/nparikh-unc/15min>

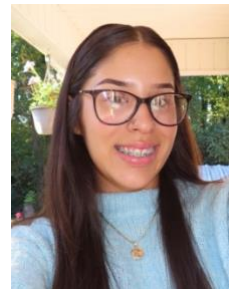
Instructional team: We are joined this year by 6 undergraduate learning assistants (ULAs) and 6 graduate student instructional assistants (IAs) who are all available to help you throughout this course!



Ann Marie Helmy, she/her
ULA - Section 2



Zoe Monroe, she/her
ULA – Section 1



Jimena Ortiz-Paniagua, she/her
ULA – Section 1



Janani Ragunathan, she/her
ULA - Section 2



Kirina Shah, she/her
ULA – Section 2



Marcellia Tyler, she/her
ULA – Section 1



Lindsay Howard, she/her
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Developmental Psyc PhD Student



Esmeralda Novarro, she/her
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Clinical Psyc PhD Student

Course Goals and Description

The primary goal of this course is for students to develop a comprehensive understanding of how research is conducted in psychological science. By the end of the course, students should understand deeply all parts of the research process, from development of a research question to execution of a research project to dissemination of scientific findings.

The course will include both a course content portion and a project portion to achieve these ambitious goals. During the content portion, we will build on your existing knowledge of the skills necessary to conduct research in psychological science. We will explore how good ideas are generated, how hypotheses are best tested, and how to think critically about psychological science. You will also be provided with opportunities for hands-on learning and to test yourselves on the skills you're learning. The project portion will provide you with an additional opportunity to practice these skills. Students will practice developing and critically evaluating research questions, analyzing data, conducting behavioral research in collaboration with their classmates, and presenting research findings effectively in both spoken and written form.

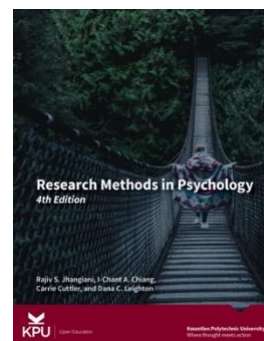
Course Outcomes

- You will be able to identify the major approaches to conducting research in psychology and understand the advantages and disadvantages of each method
- You will practice reading research analytically and writing formal APA papers.
- You will research, prepare, conduct, and report on the findings of our own quasi-experimental study.
- You will assess the validity of research claims made in academic articles.

Course Materials and Resources

Textbook: There is no required text for this course, but occasionally, I will assign a chapter from the following resources to accompany the lesson for the day. These are meant to serve as a support and resource for material that we cover in class. The resources are freely available at the following links. Throughout the course, I will also be providing handouts, other websites, and my class slides to use to guide you through your own projects.

- *Research Methods in Psychology* by Jhangiani, Chiang, Cuttler, and Leighton - <https://kpu.pressbooks.pub/psychmethods4e/>
- *Purdue University's Online Writing Lab (OWL)* for APA information - https://owl.purdue.edu/owl/research_and_citation/apa_style/apa_style_introduction.html



Online discussion space: We will be using Piazza as an online discussion space for our PSYC 270 community. This is a great space to ask course questions or fulfill your curiosities about how course material relates to life and the real world! It is also a space where other students and all instructional staff can answer any queries, so it will likely be your quickest way to get an answer. In general, rather than emailing the instructional team, I encourage you to post on Piazza. You can find our course Piazza page here: <https://piazza.com/unc/spring2025/psyc270>

Software: We will be using jamovi to analyze the data from our projects this semester. This program has been praised for its versatility and access (it's free!), and it serves as a great transition to more coding based statistical programs like R. You can download the jamovi app at [jamovi.org](https://cloud.jamovi.org/) or use a web version at <https://cloud.jamovi.org/>.



Course Assignments

Participation (15%)

Both attendance and participation are important to me, as these are ways to ensure you are keeping up with the material. But if you're worried about speaking up in a big class, fear not! There are many ways to participate in this class. The first is to complete in-class polls. (Please go to poll.unc.edu to sign up!) Polls will be scored on completeness, not correctness, and will act as a way for us to gauge how confusing class topics are! You will have 3 free, no-questions-asked skip days in case something comes up but let us know if you have a longer-term concern that keeps you from participating in class.

Secondly, you will be working in groups on your semester long research project. Each of you will be expected to contribute to the project, especially on in-class project days. At the beginning of the semester, you will all work together to form a group contract. At the end of the semester, you will be asked to assess your own contribution to the group project, and your group will also assess your contribution. Each group member will be scored on a scale of 1-10 based on the evaluations. These two assignments will count as 5% of your participation grade.

Lastly, there is a short syllabus quiz at the beginning of the semester that simply checks if you have looked through the syllabus. This quiz is open-note and will go towards your participation grade.

Learning checks (15%)

Your basic understanding of the course material will be assessed using two learning checks throughout the semester. These are meant to feel like little quizzes that are testing your knowledge in preparation for the final exam (see below).

Take-home final exam (15%)

A cumulative take-home exam will be available via Canvas. The exam will be largely multiple choice and will be open book but timed. Questions will be based on topics covered in class and through the project, and it will build on the knowledge you practiced in the learning checks.

Group research project (total of 55%)

The main goal of this course is to learn how to apply psychological research methods to your own experiments. To accomplish this goal, you will be completing a group research project where you will run a simple experiment from start to finish. (More information on the experiment will be outlined in class.) Throughout the semester, you will turn in chunks of your final paper as you go through each stage of the research process. Specifically, this project is broken down into the following components.

Article Summary and Evaluation (5%, individual grade)

Each student will select an article relevant to their final project topic. After reading this article, students will write a concise summary of it and provide a thorough critique. This assignment will help get students up to speed on existing research so that they can begin thinking about how their study fits into what has been done. In order to facilitate a broader literature review, students within a group should choose *different* articles and share their evaluations with each other.

Introduction (10%, individual grade)

Once you have a sense of the literature, you will be asked to write up an introduction that summarizes the background of your research area. How does existing work contribute to your current hypotheses?

Design a Research Study Worksheet (5%, group grade)

Research groups will work together to generate hypotheses and an experimental design for their final project. Groups will have to (i) choose which research question to pursue, (ii) consider how to operationalize their independent and dependent variables, and (iii) discuss the ethicality and appropriateness of different methods for testing their hypotheses. This is a collaborative assignment; all team members will receive the same grade.

Methods and Results (10%, group grade)

Once we have data and have looked at it, we will write up how we conducted our experiment and what we found in APA format. Fear not, we will work through the analyses and reporting in depth in class!

Final paper (25%, individual grade)

Each student will complete an APA style research paper that presents the findings of their research project. All papers must include the following sections: title page, abstract, introduction, methods, results, discussion, and references. Papers must also include at least one figure or one table depicting the study's central finding. This manuscript should be approximately 10-15 pages long (not including title page or references) and will resemble what would be submitted to an academic journal for publication. Grading will be based upon technical skill (e.g., APA style, appropriateness of study design and statistics), as well as creativity, clarity, and critical thought. More details for this paper will be given as this assignment's due date approaches.

Grading Scale

93 – 100 = A	87 – 89 = B+	77 – 79 C+	65 – 69 = D+
90 – 92 = A-	83 – 86 = B	73 – 76 C	60 – 64 = D
	80 – 82 = B-	70 – 72 C-	Below 60 = F

*All grades are rounded to the nearest whole number.

Schedule (subject to change)

Unit 1: Introduction to Research Methods

Week 1: Introduction to research methods

- Jan. 8:** **Course introduction**
Happy FWOC!
- Jan. 10:** **History of scientific psychology**
Project workshop – group formation
Form a group and begin brainstorming topics

Week 2: Ethical issues in psychological research

- Jan. 13** **Ethical issues in psychological research**
Due 1/13: Syllabus Quiz
- Jan. 15:** **Finding a topic & intro to literature searching**
- Jan. 17:** **Project workshop – topic brainstorm**
Due 1/19: Group contract

Week 3: Hypothesis Development

- ~~**Jan. 20:**~~ *No class - MLK Jr. Holiday*
- Jan. 22:** **Research questions & hypotheses**
- Jan. 24:** **Project workshop – article summary**
Begin finding articles on your topic

Week 4: Design in Psychological Research

- Jan. 27:** **Research design basics**
Due 1/27: Project part: Article summary and evaluation
- Jan. 29:** **Introduction section**
- Jan. 31:** **Project workshop – introduction**
Begin working with your group to outline the introduction section

Unit 2: Methodology

Week 5: Psychometrics

- Feb. 3:** **Reliability**
Due 2/3: Learning Check I
- Feb. 5:** **Validity**

Feb. 9: **Project workshop – introduction**
Fill in the introduction outline and complete it

Week 6: Non-experimental designs

~~**Feb. 10:**~~ *No class – wellbeing day*

Feb. 12: **Non-experimental designs & survey designs**
Due 2/13: Project part: Introduction

Feb. 14: **Project workshop – research design**
We will work on operationalizing variables, designing study flow

Week 7: Experimental and quasi-experimental designs

Feb. 17: **Experimental designs**

Feb. 19: **Quasi-experimental designs**

Feb. 21: **Project workshop – research design**
We will work on operationalizing variables, designing study flow

Week 8: Group Meetings

Feb. 24: **Project workshop – Group meetings**
The instructional team will meet with all groups individually to go over their projects
Due by 9am on Feb. 24: Project part: Design a Research Study Worksheet

Feb. 26: **Project workshop – Group meetings**

Feb. 28: **Project workshop - Group meetings**

Week 9: Other types of research designs

Mar. 3: **Factorial designs**

Mar. 5: **Methods and methods write-up**

Mar. 7: **Project workshop - data collection**
Use your fellow classmates as participants!
Due by 9am: Link to your study

Week 10: Spring break

Mar. 10-14: *No class – spring break*

Unit 3: Results and Analyses

Week 11: Intro to hypothesis testing

Mar. 17: Intro to hypothesis testing

Mar. 19 Intro to jamovi and data cleaning

Mar. 21: Project workshop – running analyses
Run analyses and work on your results section
Due on 3/21: Learning Check II

Week 12: Analysis plan and write up

Mar. 24: Choosing the right analysis

Mar. 26 Results in APA format

Mar. 28: Project workshop – results write-up
Due 4/1: Project part: Methods and Results

Unit 4: Discussion

Week 13: Issues impacting validity

Mar. 31: Threats to validity

Apr. 2: Threats to validity II

Apr. 4: Project workshop – discussion
Make substantive interpretations of the project results

Week 14: Final paper prep

Apr. 7: Threats to validity review

Apr. 9: APA formatting

Apr. 11: Project workshop – discussion
Make substantive interpretations of the project results

Week 14: Final paper prep

Apr. 14: Replication crisis

Apr. 16: Project workshop – discussion and final paper
Complete the discussion section and work on formatting
Due: Project part: Group dynamics

~~**Apr. 18:**~~ No class – University holiday

Week 16: Review

- Apr. 21:** **No class**
Work on your final paper / take a breather!
- Apr. 23:** **Project workshop – Last final paper questions**
Due: Project part: Final paper
- Apr. 25:** **Overview and final exam review**

Finals

- Apr. 28:** **LDOC – TBD**
- Apr. 30:** **Final exam due**

Course Policies

Attendance and late assignments

University Policy: As stated in the University's [Class Attendance Policy](#), no right or privilege exists that permits a student to be absent from any class meetings, except for these University Approved Absences:

1. Authorized University activities
2. Disability/religious observance/pregnancy, as required by law and approved by [Accessibility Resources and Service](#) and/or the [Equal Opportunity and Compliance Office](#) (EOC)
3. Significant health condition and/or personal/family emergency as approved by the [Office of the Dean of Students](#), [Gender Violence Service Coordinators](#), and/or the [Equal Opportunity and Compliance Office](#) (EOC).

If you anticipate that you will miss class for any of the above, please let me know as soon as possible (and at least 48 hours before your conflict) so I can arrange make up material for you. Otherwise, your absence will come from your 3 free skip days. If you have a contagious illness or an emergency, please let me know ASAP.

Late policy: Late project assignments will be docked 2% per day. Late learning checks/exams will be docked 10% per day late.

Honor Code and AI Policy

I expect all students to follow the guidelines of the UNC Honor Code. In particular, students are expected to refrain from "lying, cheating, or stealing" in the academic context. You can read more about the honor code at studentconduct.unc.edu. In any course, including mine, what constitutes cheating can change from one activity to another. For example, collaboration may be encouraged for a writing assignment but qualify as cheating during an exam. Please see my guidelines for each activity, and if you are unsure, please ask me to clarify. Please don't get caught up with honor code issues just because it appears to be simple and untraceable. It is not!

For this course, using generative AI to write any aspect of your research paper would be considered a form of cheating. The goal of this course is to teach you how to write like a scientist, so if a computer does this work for you, you are plagiarizing the work of an online source. However, generative AI can be helpful (and is allowed) for small tasks such as: fine-tuning your research statement, suggesting research papers in your topic area, and finding commonalities between the articles you have found. In these cases, it is best practice to cite where you used generative AI as a footnote to your assignment. Remember, AI is often wrong, so make sure to check the output for correctness!

Computer use

I expect that computer use will be limited to in-class activities. Please do not access social media, online shop, or text during course time. These activities not only keep you from learning the material and participating in class; they also distract others around you and lessen their participation. If I notice repeated non-class technology activity, I will reduce your lab grade for that class period. However, it can be hard to stay focused all the time. If you know you are expecting a call or need to be available for something urgent, please let me know. I understand that sometimes life requires you to be more plugged in!

Policy on Non-Discrimination

The University is committed to providing an inclusive and welcoming environment for all members of our community and to ensuring that educational and employment decisions are based on individuals' abilities and qualifications. Consistent with this principle and applicable laws, the University's [Policy Statement on Non-Discrimination](#) offers access to its educational programs and activities as well as employment terms and conditions without respect to race, color, gender, national origin, age, religion, genetic information, disability, veteran's status, sexual orientation, gender identity or gender expression. Such a policy ensures that only relevant factors are considered, and that equitable and consistent standards of conduct and performance are applied.

If you are experiencing harassment or discrimination, you can seek assistance and file a report through the Report and Response Coordinators (see contact info at safe.unc.edu) or the [Equal Opportunity and Compliance Office](#), or online to the EOC at <https://eoc.unc.edu/report-an-incident/>.

Diversity and Inclusion in the Classroom

I view the diversity represented by students in this class as a strength. It is our intent that students from diverse backgrounds and perspectives be well-served by this course and that all students' learning needs be addressed both in and out of class, and that the diversity that the students bring to this class be viewed as a resource, strength, and benefit. I aim to teach in ways that acknowledge and celebrate diversity in gender identity, sexuality, disability, age, socioeconomic status, ethnicity, race, nationality, religion, and culture. I invite you to help me identify and recognize things I have overlooked or do not yet know: your suggestions are encouraged and appreciated.

Basic Needs

Any student who has difficulty affording groceries or accessing sufficient food to eat every day or lacks a safe and stable place to live and believes that this may affect their performance in the course, is urged to contact the Dean of Students for support. Furthermore, they can talk to me if they feel comfortable, so I can also provide any additional resources I may have access to.

Accommodations

The University of North Carolina at Chapel Hill facilitates the implementation of reasonable accommodations, including resources and services, for students with disabilities, including mental health disorders, chronic medical conditions, a temporary disability or pregnancy complications resulting in barriers to fully accessing University courses, programs and activities.

Accommodations are determined through the Office of Accessibility Resources and Service (ARS) for individuals with documented qualifying disabilities in accordance with applicable state and federal laws. See the ARS Website for contact information: <https://ars.unc.edu> or email ars@unc.edu. Please also email me within the first week of class to let me know.

Counseling and Psychological Services

UNC-Chapel Hill is strongly committed to addressing the mental health needs of a diverse student body. The [Heels Care Network](#) website is a place to access the many mental resources at Carolina. CAPS is the primary mental health provider for students, offering timely access to consultation and connection to clinically appropriate services. Go to their website <https://caps.unc.edu/> or visit their facilities on the

third floor of the Campus Health building for an initial evaluation to learn more. Students can also call CAPS 24/7 at 919-966-3658 for immediate assistance.

Title IX Resources

Any student who is impacted by discrimination, harassment, interpersonal (relationship) violence, sexual violence, sexual exploitation, or stalking is encouraged to seek resources on campus or in the community. Reports can be made online to the EOC at <https://eoc.unc.edu/report-an-incident/>. Please contact the University's Title IX Coordinator (Elizabeth Hall, titleixcoordinator@unc.edu), Report and Response Coordinators in the Equal Opportunity and Compliance Office (reportandresponse@unc.edu), Counseling and Psychological Services (confidential), or the Gender Violence Services Coordinators (gvsc@unc.edu; confidential) to discuss your specific needs. Additional resources are available at safe.unc.edu.

Undergraduate Testing Center

The College of Arts and Sciences provides a secure, proctored environment in which exams can be taken. The center works with instructors to proctor exams for their undergraduate students who are not registered with ARS and who do not need testing accommodations as provided by ARS. In other words, the Center provides a proctored testing environment for students who are unable to take an exam at the normally scheduled time (with pre-arrangement by your instructor). For more information, visit <http://testingcenter.web.unc.edu/>.

Learning Center

Want to get the most out of this course or others this semester? Visit UNC's Learning Center at <http://learningcenter.unc.edu> to make an appointment or register for an event. Their free, popular programs will help you optimize your academic performance. Try academic coaching, peer tutoring, STEM support, ADHD/LD services, workshops and study camps, or review tips and tools available on the website.

Writing Center

For free feedback on any course writing projects, check out UNC's Writing Center. Writing Center coaches can assist with any writing project, including multimedia projects and application essays, at any stage of the writing process. You don't even need a draft to come visit. To schedule a 45-minute appointment, review quick tips, or request written feedback online, visit <http://writingcenter.unc.edu>.

Grade Appeal Process

If you feel you have been awarded an incorrect grade, please discuss with me. If we cannot resolve the issue, you may talk to our departmental director of undergraduate studies or appeal the grade through a formal university process based on arithmetic/clerical error, arbitrariness, discrimination, harassment, or personal malice. To learn more, go to the [Academic Advising Program](#) website.

IDEAS in Action Learning Outcomes

General Learning Objectives:

1. Writing, totaling at least 10 pages in length or the intellectual equivalent
In this course, we will fulfill an equivalent of 10 pages through our research paper.

2. Presenting material to the class, small groups, or the public through oral presentations, webpages, or other means
We will present our work to the instructional team in small groups.
3. Collaborating in pairs or groups to learn, design, solve, create, build, or research.
Your research project will be done with a group of your peers. Some of the paper components are written as a team.

Empirical Investigation Lab Focus Capacity: Students will create experiments, collect data, and analyze the results to further their abilities to investigate the world around them.

1. Take empirical measurements using appropriate apparatus.
We will use Qualtrics to gather data.
2. Generate and test hypotheses.
As part of the research design, you will make hypotheses of what you expect to occur.
3. Gather, store, and organize data.
We will collect data from our classmates as well as those outside of class.
4. Analyze and report on data and hypothesis testing.
At the end of the semester, we will present our findings in our final paper.

Research and Discovery (Reflection & Integration): Students will propose original research ideas and present them to their peers.

1. Frame a topic, develop an original research question or creative goal, and establish a point of view, creative approach, or hypothesis.
You will come up with an original research question as you develop your research project.
2. Obtain a procedural understanding of how conclusions can be reached in a field and gather appropriate evidence.
We will discuss statistical analyses methods and collect our own data to analyze.
3. Evaluate the quality of the arguments and/or evidence in support of the emerging product.
We will evaluate whether our studies worked as expected given the analyses that we undertake.
4. Communicate findings in a clear and compelling ways.
We will present our findings by writing a full, scientific research paper!
5. Critique and identify the limits of the conclusions of the project and generate ideas for future work.
In our discussion section, we will work through limitations and future directions.